Comparative Analysis Report

Baseline vs Naive RAG (Solution V1) vs Hybrid RAG (Solution V2)

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1. Executive summary

This report compares three progressively enhanced customer-support systems built on Qwen2.5-1.5B-Instruct (4-bit): a Baseline pre-trained model, a Naive RAG system (Solution V1) that retrieves a corporate SOP using the raw customer query, and a Hybrid RAG system (Solution V2) in which a LoRA fine-tuned intent router turns the noisy query into a structured JSON intent that drives retrieval. All three were scored on the full 391-row held-out test split against SOP-grounded references.

- **Fine-tuning gave the clearest gain.** Exact intent match rose from 1.0% (V1) to 34.0% (V2) and fuzzy match from 19.9% to 67.5%. The Baseline scored 1.8% exact and 7.7% fuzzy.
- **Retrieval alone did not improve intent labels.** On the full test split V1 scored 1.0% exact against 1.8% for the Baseline. In the live 50-row V1 evaluation, intent accuracy fell from 2% to 0% while category accuracy rose from 6% to 10%.
- **Format adherence is high but not perfect.** V2 returned valid JSON in 93.1% of final responses and 98.2% of router outputs (V1: 94.6%, Baseline: 100%).
- **ROUGE, BLEU and hallucination do not separate the systems.** The systems emit short JSON labels and the references are prose SOPs, so ROUGE-1 stays between 0.014 and 0.026 and BLEU is 0.000 throughout. Hybrid's hallucination rate is 1.0% (4 of 391 rows).
- **Targets.** Two of the seven proposal targets were met (format adherence and consistency). The ROUGE and BLEU targets assumed prose answers and were not reachable with short JSON outputs.

2. Experimental setup

System	Model and precision	Retrieval	Output
Baseline	Qwen2.5-1.5B-Instruct, 4-bit NF4	None	Intent and category JSON from a zero-shot system prompt
Solution V1 Naive RAG	Same model	all-MiniLM-L6-v2 + ChromaDB; top-1 SOP found with the raw query and injected into the system prompt	Intent and category JSON
Solution V2 Hybrid RAG	Same model with the LoRA adapter merged (router)	Top-1 SOP found with a search string built from the router's intent (for example track_order becomes “Track Order”); raw query if the router JSON is invalid	Final answer generated with the SOP as context

Evaluation protocol

- **Data.** Bitext sample of 4,000 rows (seed 42), 3,902 after normalised deduplication, split 80/10/10 stratified by intent into 3,121 / 390 / 391 rows. Set-intersection checks confirmed zero overlap between the splits.
- **Comparison set.** All 391 test rows, identical for all three systems. The live V1 evaluation in Notebook 5 uses a 50-row sample (seed 42).
- **Adversarial subset.** The workflow's suggested words matched none of the test rows, so the subset uses words extracted from the data (again, problem, issue, wrong, help, cannot, can't): 101 rows matched and all were evaluated, on V2 only.
- **Decoding.** Greedy for every system (do_sample=False, temperature=None, top_p=None).
- **Metrics.** Format adherence: the output parses as JSON (for V2, the final response, with router adherence reported separately). Exact intent match: predicted intent equals the Bitext label. Fuzzy intent match: at least half of the true label's words appear in the prediction. ROUGE-1, ROUGE-L and BLEU: against the top SOP for the raw query. Hallucination: a response is flagged if it contains

a timeline, “department X”, “call us at” or “tracking number X” that the reference SOP does not contain. Consistency: identical output across 3 repeated runs on 5 queries.

- **Provenance.** Baseline and V1 outputs come from Notebooks 3 and 4 (saved in outputs.json, 4-bit); V2 was generated live in Notebook 7. Consistency for Baseline and V1 was measured in Notebook 5 and for V2 in Notebook 7.

3. Results (Task 4.5.1)

3.1 All systems, all metrics

Metric (391 test rows)	Baseline	Solution V1 Naive RAG	Solution V2 Hybrid RAG
Format adherence (%)	100.0	94.6	93.1
Exact intent match (%)	1.8	1.0	34.0
Fuzzy intent match (%)	7.7	19.9	67.5
ROUGE-1	0.014	0.026	0.021
ROUGE-L	0.014	0.023	0.018
BLEU	0.000	0.000	0.000
Hallucination frequency (%)	0.0	0.0	1.0
Consistency (%)	100.0	100.0	100.0

Table 1. Final comparison. One row equals 0.26 percentage points. V2 router-only format adherence is 98.2%. Consistency uses 5 queries × 3 runs.

3.2 Retrieval impact: Baseline to Solution V1

Metric (Notebook 5, live, 50 rows)	Baseline	Naive RAG	Change
Format adherence (%)	100.0	100.0	0.0%
Intent accuracy (%)	2.0	0.0	−100.0%
Category accuracy (%)	6.0	10.0	+66.7%
ROUGE-1	0.016	0.029	+81.2%
ROUGE-L	0.014	0.026	+85.7%
BLEU	0.000	0.000	n/a
Consistency (%)	100.0	100.0	0.0%
Hallucination (%)	0.0	0.0	n/a

Table 2. Notebook 5 generates live with 120 new tokens and a strict JSON schema check on a 50-row sample, so its V1 figures differ from the saved-output column in Table 1 (for example format adherence 100% against 94.6%). “n/a” means the baseline was zero.

3.3 Fine-tuning impact: Solution V1 to Solution V2

Metric (391 rows)	Solution V1	Solution V2	Change
Format adherence (%)	94.6	93.1	−1.6%
Exact intent match (%)	1.0	34.0	+33.0 points
Fuzzy intent match (%)	19.9	67.5	+47.6 points
ROUGE-1	0.026	0.021	−15.5%
ROUGE-L	0.023	0.018	−17.7%
BLEU	0.000	0.000	0.0%
Hallucination (%)	0.0	1.0	+1.0 points

Table 3. Percentages for format adherence, ROUGE and BLEU are as printed in Notebook 7 (Task 4.4.2); the fuzzy-match and hallucination changes are computed from Table 1.

3.4 Adversarial subset (Solution V2 only)

Metric	Full test split (391 rows)	Adversarial subset (101 rows)
Format adherence (%)	93.1 (364/391)	95.0 (96/101)

Metric	Full test split (391 rows)	Adversarial subset (101 rows)
Router format adherence (%)	98.2	98.0
Exact intent match (%)	34.0 (133/391)	33.7 (34/101)
Fuzzy intent match (%)	67.5 (264/391)	69.3 (70/101)
ROUGE-1 / ROUGE-L / BLEU	0.021 / 0.018 / 0.000	0.019 / 0.017 / 0.000

Table 4. Accuracy on the harder queries is essentially the same as on the full split; the router is not noticeably more fragile on hedged or difficult phrasing.

3.5 Training curves

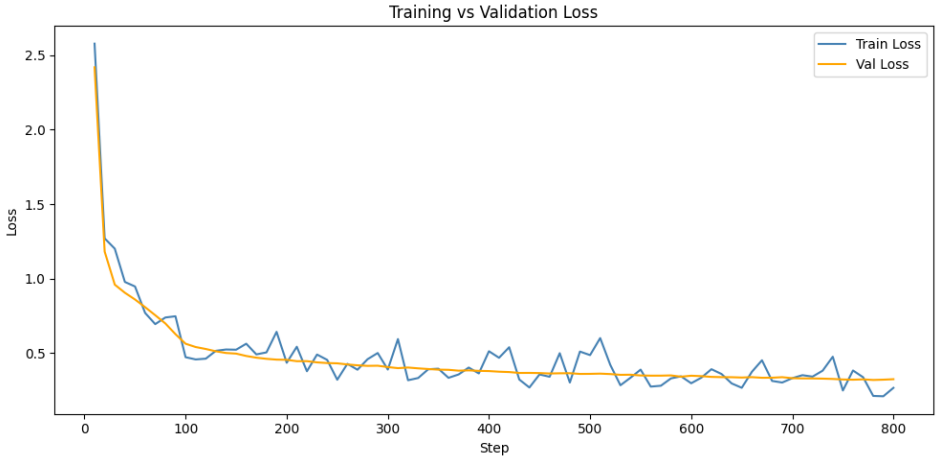


Figure 1. Training and validation loss of the router over 800 steps, from the fine-tuning notebook. Validation loss fell from 2.42 at step 10 to a best of 0.320 at step 780; early stopping never triggered.

3.6 Example end-to-end runs (Task 4.3.1 validation)

Query	Router output	Retrieved SOP	Final response
i ordered something 2 weeks ago and it still hasn't arrived!	{"intent": "track_order", "category": "order_status"}	order_tracking.md	{"intent": "delivery_status", "category": "order_tracking"}
i need a refund for my order	{"intent": "refund", "category": "payment"}	refund_policy.md	{"intent": "refund", "category": "refunding"}
how do i reset my password?	{"intent": "set_password", "category": "account"}	password_reset.md	Plain text: “click here to reset your password: [link]”

Table 5. The router retrieves the right SOP in all three cases. Only the first intent (track_order) is a Bitext label; the others (refund, set_password) are close but off-schema, the categories are lowercase and off-vocabulary, and the final stage returns JSON in two cases and free text in one.

4. Why Hybrid RAG shows a hallucination rate

Hybrid RAG is the only system with a non-zero reading (1.0%, 4 of 391 rows). Baseline and V1 output only short labels, which cannot contain the phrases the checker looks for, while V2's final stage sometimes writes free text. The checker compares each response with the SOP retrieved for the raw query, whereas Hybrid answers from the SOP retrieved for the router's intent, so a response can be flagged for wording that is faithful to the SOP it was actually given. The figure is therefore best read as a measure of retrieval disagreement and router error rather than of fabricated policy, and with four flagged rows it is a small-sample number.

5. Key findings (Task 4.5.2)

- **1. Fine-tuning is the main contributor.** From V1 to V2, exact match rises from 1.0% to 34.0% and fuzzy match from 19.9% to 67.5% on 391 rows. The gap is large and not within sampling noise.
- **2. Retrieval alone does not fix intent labels.** Neither the Baseline nor V1 is told the 27-label vocabulary, so both produce plausible but off-schema labels such as “customer_service” and “order tracking”. The retrieved SOP supplies topic context, not label vocabulary. In the live 50-row

run, category accuracy improved (6% to 10%) and ROUGE roughly doubled from a very low base.

- **3. The router is still short of the exact-match target.** 34.0% exact against an 80% target. Fuzzy match is lenient (at least half of the true label's words), so the 67.5% figure shows the router often lands on the right topic, not that it reproduces the label set; its categories remain lowercase and partly off-vocabulary.
- **4. ROUGE, BLEU and hallucination are weak evidence here.** The systems emit short JSON while the references are prose, so ROUGE-1 ranges only from 0.014 to 0.026 and BLEU is 0.000 for all three. The drop in ROUGE from V1 to V2 should not be read as evidence against fine-tuning.
- **5. The final stage is not format-stable.** Although the prompt asks for JSON only, the final answer fails the JSON check on 6.9% of rows (93.1% adherence), against 1.8% for the router (98.2%).
- **6. The router holds up on harder queries.** Exact match is 33.7% on the 101-row adversarial subset against 34.0% on the full split.

Targets set at the proposal stage versus results

Metric	Target	Achieved	Status
Format adherence	≥ 85%	93.1% (V2 final response, 364/391); router 98.2%	Met
Intent accuracy	≥ 80%	34.0% exact / 67.5% fuzzy	Not met
ROUGE-1	≥ 0.35	0.021	Not met
ROUGE-L	≥ 0.30	0.018	Not met
BLEU	≥ 0.20	0.000	Not met
Consistency	≥ 90%	100% (5 queries × 3 runs)	Met
Hallucination reduction	≥ 30% vs Baseline	Baseline 0% → V2 1.0%	Not met

Table 6. The ROUGE and BLEU targets assumed prose answers; because the systems emit short JSON labels, they could not be reached by design. The hallucination target could not be met because the Baseline was already at 0%.

6. Limitations

- **1. Output design.** All systems return a short classification JSON rather than a customer-facing answer, so the retrieved SOP seldom influences the output and the SOP-based metrics carry little information.
- **2. Evaluation samples.** The main comparison uses the full 391-row test split, but the adversarial subset has 101 rows, the live V1 evaluation (Notebook 5) uses only 50 rows, and consistency is measured on 5 queries.
- **3. Short, unseeded training.** Fine-tuning ran for 800 steps (about one epoch) with no training seed. Validation loss reached 0.320 and was still improving slowly, so the router is probably under-trained; run-to-run variance was not measured, and the reported V2 figures are one draw from a noisy process.
- **4. Loss masking departs from the notebook text.** The loss covers only the JSON answer (padding and all prompt tokens are masked with −100), whereas the Task 2.3.1 markdown in the data-preparation notebook describes padding-only masking. The task text was not updated, so the notebook and this report differ on purpose.
- **5. Overlapping labels.** Several Bitext intents are close in meaning (for example get_refund and check_refund_policy), which caps exact-match accuracy for a small model trained for about one epoch.
- **6. Hallucination checker.** It uses four regular-expression patterns and compares against the SOP for the raw query, so it can flag answers that are faithful to the SOP actually retrieved (Section 4).
- **7. Data characteristics.** About 23% of sampled prompts contain template placeholders such as {{Order Number}}, and some contain typos. The 13 SOP files are stored as whole documents without chunking.
- **8. Merged adapters.** Merging LoRA weights into 4-bit layers can change generations slightly because of rounding.

- **9. Two V1 figures.** V1 was scored live on 50 rows in Notebook 5 and from saved outputs on 391 rows in Notebook 7; format adherence is 100% in one and 94.6% in the other. The final comparison uses the saved outputs so that all three systems are scored identically.

7. Recommendations and future work

- **1.** Train for more than one epoch (for example about 1,500 steps at batch size 4) with a fixed seed, and watch whether validation loss keeps falling; if it flattens, try a higher LoRA rank or add k_proj and o_proj.
- **2.** Constrain the router's output to the 27 valid intents (and 11 uppercase categories), for example with constrained decoding or a post-hoc mapping.
- **3.** Replace title-cased search strings with a deterministic intent-to-SOP table (for example track_order to order_tracking.md), keeping semantic search as a fallback.
- **4.** Add a free-text, SOP-grounded answer stage and evaluate faithfulness against the SOP actually retrieved, plus a retrieval hit-rate against a gold intent-to-SOP map.
- **5.** Fix the format drift in the final stage with a separate answer prompt or a constrained JSON response.
- **6.** Repeat the evaluation across several seeds and report confidence intervals; run the live V1 evaluation on the full test split.
- **Future work.** A larger base model (for example Qwen2.5-3B or Llama-3-8B with QLoRA) if compute allows; chunked SOPs with hybrid keyword plus dense retrieval; human review of generated answers and multi-turn conversations.

8. Conclusion

Fine-tuning an intent router is what moved the system: exact intent match went from 1.0% to 34.0% and fuzzy match from 19.9% to 67.5% on the full test split, while retrieval alone left intent accuracy unchanged. The remaining gaps are exact label accuracy (34%, against an 80% target), format drift in the final answer stage (93.1% adherence), the lack of a free-text answer stage that would let the SOP matter, and metrics that do not discriminate between JSON outputs. The recommendations above address these in order of expected impact.

Appendix A. Configuration summary

Item	Setting
Base model	Qwen/Qwen2.5-1.5B-Instruct, 4-bit NF4 (double quantisation, fp16 compute) for all systems
Decoding	do_sample=False, temperature=None, top_p=None; max_new_tokens 30 (router) and 120 (final answer in Notebook 7, live generation in Notebooks 3 and 5)
Retrieval	sentence-transformers/all-MiniLM-L6-v2 (384-d), ChromaDB, 13 SOP documents, top-1
LoRA	r=16, alpha=32, dropout 0.05, target modules q_proj and v_proj; 2,179,072 trainable parameters of 1,545,893,376 (0.141%)
Training	Custom AdamW loop, learning rate 2e-4 (constant), batch size 4, 800 steps, validation every 10 steps on all 390 validation rows, early-stopping patience 3, best checkpoint by validation loss; loss on the JSON answer only (padding and prompt tokens masked)
Training outcome	Validation loss 2.42 (step 10) to a best of 0.3203 (step 780), 0.3254 at step 800; early stopping not triggered; training seed not set
Data	26,872 rows, 4,000 sampled (seed 42), 3,902 after deduplication; stratified split 3,121 / 390 / 391 (seed 42)
Environment	Stages 1–2 on a local Jupyter notebook; Stages 3–4 on Kaggle (T4 GPU)